

Savora AI

Smart Food Recognition and Calorie Estimation System

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1. Introduction

The importance of healthy living has grown more critical in recent years because students and working professionals face continuous challenges from their unpredictable work schedules. People have become more aware of their daily eating habits because of increased public knowledge about fitness and nutrition and overall health.

People find it hard to track their diet because traditional tracking methods need them to enter data manually and keep tracking their food intake and they must learn about food nutritional information. People have to spend time and effort to use manual calorie tracking applications which leads to them using the applications less frequently. People find it difficult to track their meals because they have to do portion size estimation and calorie value calculation manually. People require a solution which can handle the process through automated operations and efficient functions while providing simple user interaction.

Artificial intelligence combined with deep learning has made new solutions possible because these technologies enable scientists to use them for advanced image processing methods. The effectiveness of Convolutional Neural Networks (CNNs) in image classification enables them to serve as food identification systems which can recognize food products from visual data. The development of food recognition systems which require only minimal user intervention to determine food items and their nutritional information became possible through these technological advancements.

The project aims to create a web-based platform which can analyze food images to identify different food items and calculate their calorie values. The system uses a deep learning model which was trained to identify different food items through food image classification while the nutrition database provides approximate calorie estimates based on standard serving sizes. The user simply needs to upload an image, and the system processes the image to create predictions together with corresponding dietary details.

The main goal of this project is to develop a diet tracking system which operates with high speed and direct user access and complete efficiency without requiring users to input data manually. The system applies artificial intelligence technology through its user-friendly design to generate user-friendly results which help people track their food intake more accurately while creating better eating habits. The system provides time savings which enable more users to conduct nutritional monitoring through its simpler design.

2. Literature Survey

Deep learning techniques have been applied in multiple studies to recognize food from different samples. Convolutional Neural Networks (CNNs) have been widely used for image classification tasks

because they can extract spatial features from images with high efficiency. Food tracking applications need users to enter information manually which results in user fatigue and creates problems with data accuracy. The research papers demonstrate that automated food recognition systems enhance both user engagement levels and accurate results. The majority of systems lack the capacity to estimate food portions while they also fail to provide real-time operational capabilities.

Some models achieve high accuracy but they need both extensive data sets and powerful computational resources to operate successfully. The project aims to achieve accurate results while maintaining operational efficiency through its use of a moderate data set and its lightweight deployment method which is designed for web-based applications.

3. Dataset Description

The project used a dataset which includes multiple food categories. The food categories used in this project include Pizza, Biryani, Poha, Idli, Cake and Others. Each category contains approximately 100+ images collected from various sources. The researchers used labeled folders to organize images which helped them conduct supervised learning. The dataset underwent preprocessing before training began. Images were resized to a fixed dimension. The researchers applied data augmentation techniques which included rotation and flipping to the images. The researchers divided the dataset into two parts which included training and validation sets. The process enabled models to generalize better while simultaneously decreasing overfitting problems.

4. Methodology

The system employs a hybrid method which integrates deep learning with rule-based estimation techniques.

Step 1: Image Input

The user uploads an image through the web interface.

Step 2: Image Preprocessing

The uploaded image is resized and normalized to match the input format required by the model.

Step 3: Model Prediction

The food item is classified by a Convolutional Neural Network (CNN) model which generates two outputs. The model outputs:

- Predicted class label
- Confidence score

Step 4: Top Predictions

The system shows its three most probable predictions along with their certainty values to provide better visibility.

Step 5: Calorie Estimation

The system predicts food items which leads to calorie estimation based on specific portion sizes from the existing nutrition database.

Step 6: Output Display

The result includes:

- Predicted food name
- Confidence score
- Estimated calories
- Portion size

5. Results and Discussion

The model demonstrated effective performance for trained food categories which included pizza and biryani and idli. The accuracy results showed significant improvement through dataset expansion and the application of data augmentation techniques.

Observations:

- The system achieves high accuracy when foods can be easily identified.
- The system confuses visually similar items such as poha and upma.
- The system uses confidence scores to help identify uncertain prediction results.

Challenges:

- The available dataset contains insufficient sample data.
- The study faces challenges due to different lighting conditions and varying image quality.
- The study involves food items that share similar visual characteristics.

Improvements Made:

- The dataset now contains more than 100 images for each specific class.
- The system now provides top-three prediction results.
- The user interface received updates which enhanced its text readability.

Future Scope:

- The system will introduce additional food categories for identification.
- The system will use object detection to enable portion size identification.
- The system will require larger datasets to achieve improved accuracy.
- The system will receive camera input from real-time systems for processing.

Conclusion

The project demonstrates that deep learning methods can be used to solve actual problems in food identification and calorie counting. Image classification and nutrition-based estimation system integration enable users to track their daily food consumption through automatic tracking.

The system enables users to upload food images for prediction results which include calorie estimation on a user-friendly and efficient platform. People find it easier to make better dietary decisions through this system which provides them with better access to information. The system uses confidence scores and top predictions to show users which results can be trusted more.

The development process saw multiple improvements through dataset expansion and data augmentation and interface enhancements which improved both accuracy and usability. The system can still operate successfully in trained categories despite existing challenges from food items that look alike and different image conditions.

This project demonstrates how deep learning technology can be applied to develop online health and nutritional tools. The system will become a more powerful and expandable solution through future improvements which include bigger datasets and real-time image analysis and portion size identification. The system will develop into an all-encompassing application that helps users achieve their health goals through better daily practices.